

EXPERIMENT 22. Reference Resolution

Aim: To create a Python program that performs simple reference resolution within a text.

Program:

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = "John went to the store. He bought a book. It was expensive."

doc = nlp(text)

last_noun = None

print("Reference Resolution:\n")

for token in doc:

    if token.pos_ in ["PROPN", "NOUN"]:

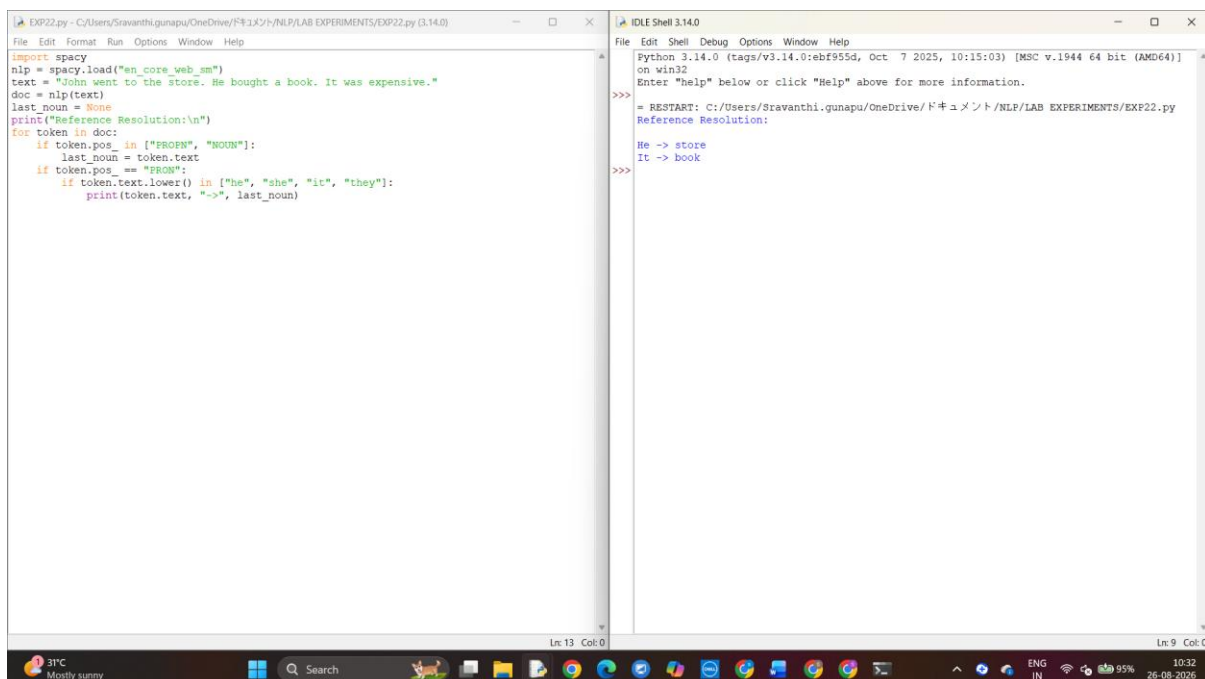
        last_noun = token.text

    if token.pos_ == "PRON":

        if token.text.lower() in ["he", "she", "it", "they"]:

            print(token.text, "->", last_noun)
```

Output:



The screenshot shows a Python IDE with two windows. The left window displays the Python code for reference resolution, and the right window shows the output of the program.

```
EXP22.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP22.py (3.14.0)
File Edit Format Run Options Window Help
import spacy
nlp = spacy.load("en_core_web_sm")
text = "John went to the store. He bought a book. It was expensive."
doc = nlp(text)
last_noun = None
print("Reference Resolution:\n")
for token in doc:
    if token.pos_ in ["PROPN", "NOUN"]:
        last_noun = token.text
    if token.pos_ == "PRON":
        if token.text.lower() in ["he", "she", "it", "they"]:
            print(token.text, "->", last_noun)
Ln: 13 Col: 0
```

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP22.py
Reference Resolution:
He -> store
It -> book
>>>
Ln: 9 Col: 0
```

The taskbar at the bottom shows the system clock as 10:32 on 26-08-2025, with a battery level of 95% and network status.